

CMPSC_431W

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PHASE-2 SYSTEM PROTOTYPE

INSTRUCTIONS TO DEPLOY SYSTEM

HOW TO CREATE DATABASE "KURALDB1"

Step 1] Please verify the following details :

(a)Details of Database DUMP.

(b)Dump name:

Database Name: KURALDB1

Database Type: MySQL

Server version: 8.0.23

TARGET_DIR=/any_main_folder/<dump_dir> example : /home/enduser/kural/sql/hw4/

Step 2] Please copy the dump file "KURALDB1_db_dump.sql" into TARGET_DIR

```
##cp KURALDB1_db_dump.sql $TARGET_DIR
```

```
##Example: cp KURALDB1_db_dump.sql /home/enduser/kural/sql/hw4/
```

Step 3] Please do the following to import all Tables/views and Tables Data.

```
#Instruction to import Database "KURALDB1"
```

```
cd $TARGET_DIR
```

```
mysql -u root -p KURALDB1 < ./KURALDB1_db_dump.sql
```

step 4] SELECT customer_data table data to confirm Import is successful.

```
mysql -h localhost -u root -p
```

```
use KURALDB1 ;
```

```
select * from customer_data ;
```

```
-----  
-----
```

HOW TO INSTALL KURAL BOOKSTORE APPLICATION

Step 1] Please verify the following:

Application Tar File name: kuralhw4_20210423.tar.gz

Application written in: Django

Application TARGET_DIR=/any_main_folder/<dump_dir> example : /home/enduser/kural/sql/hw4

Step 2] Please copy and extract tar file of source code in source dir

```
##cd TARGET_DIR
```

Example: cd /home/enduser/kural/sql/hw4

```
tar -xvzf kuralhw4_20210423.tar.gz .
```

Step 3] This source code already have python virtual environment. goto "bs" dir and please source it.

```
##cd TARGET_DIR/bs
```

Example :cd /home/enduser/kural/sql/hw4/bs

```
source bin/activate
```

Step 4] Please make sure you have the following packages has been installed in virtual environments.

```
pip3 freeze
```

```
asgiref==3.3.4
```

```
Django==3.2
```

```
django-widget-tweaks==1.4.8
```

```
mysqlclient==2.0.3
```

```
pytz==2021.1
```

```
sqlparse==0.4.1
```

```
typing-extensions==3.7.4.3
```

Step 5] Please Verify Django and mysqlclient package installed (listed as above).

IMPORTANT NOTE: Without Django and mysqlclient package this application won't work.

Step 6] Verify MySQL DB connection.

To connect to MySQL DB from Application I have configured in settings.py like the following , please modify as per your infrastructure.

```
'default': {  
    'NAME': 'KURALDB1',  
    'ENGINE': 'django.db.backends.mysql',  
    'USER': 'root',  
    'PASSWORD': 'Mahendran#14',  
    'HOST': '127.0.0.1',  
    'OPTIONS': {  
        'autocommit': True,  
    },  
}
```

Step 7] To Start the application, go to application main directory.

```
##cd TARGET_DIR/bs/src
```

Example : cd /home/enduser/kural/sql/hw4/bs/src

NOTE: To start application in https mode, i have created certification and i have implemented in "stunnel" Directory

Step 8] To Start the application, do the following:

```
./runserver
```

Step 9] Following information i have pre configured to access the application

MANAGER ID /PWD = kural200/mahendran

USER ID /PWD = kural002/mahendran

Step 10] Use this URL: to access the application.

http://127.0.0.1:8000/

FUNCTIONALITIES DESCRIPTION:

When the system is deployed, the first screen you can view is the login page with manager login, login and register options as functionalities.

The register functionality allows Customers to register themselves. The Customer would have to fill in basic information according to the constraints as shown below:

Kuraloviyan Book Store

Manager LoginLoginRegister

User Registration

Login ID:

First name:

Last name:

Address:

Mobile number:

Email id:

User password:

Register

The constraints to registration are:

- (a) If the Login ID already exists, the system would ask to try another login ID.
- (b) Mobile Number would only accept numbers and no other characters.
- (c) Email ID needs to contain the @ character.
- (d) The password length should not be less than 8 characters in length, and it is hashed while typing.
- (e) No field can be left blank.

User Registration

Login ID: First name: Last name:

Address:

Mobile number: Email id:

- Password length should not be less than 8 characters

User password:

Once an account is created, a customer can use to the login functionality to login to their account as shown below:

User Login

User Login ID: User password:

If a manager wishes to login to their account, they can use the manager login option.

Only a manager can add another manager and a manager cannot register themselves:

Kuraloviyan Book Store

Manager LoginLoginRegister

Manager Login

User Login ID:

User password:

Manager LoginCancel

Once logged in, the customer can use many functionalities as shown below:

Kuraloviyan Book Store ▾

SearchBookSearchCommentsPrev OrdersRecommPurchaseCustProfileLogout

In the SearchBook functionality, the customer can search for a book title or an author or a particular publisher or even a favorite subject:

Kuraloviyan Book Store ▾

SearchBookPrev OrdersRecommPurchaseCustProfileLogout

Search Page

Search Book title or Author or Publisher or Subject

For example, if a customer wishes to search for a Python book, we will then see a list of the book details related to Python books and below the screen we find an option to search books again. The book details which can be viewed can be seen below:

Kuraloviyan Book Store

SearchBook

Prev Orders

Recomm

Purchase

CustProfile

Logout

Book Details

isbn	Book Title	Author name	Publisher name	Language	Price
0312311451	The Python's Autobiography by The Python's	Graham Chapman/John Cleese/Terry Gilliam/Eric Idle/Terry Jon	St. Martin's Griffin	eng	100.00
0679726470	The Complete Monty Python's Flying Circus: All the	Graham Chapman/Eric Idle/Terry Gilliam/John Cleese/Terry Jon	Pantheon	eng	100.00
0679726489	The Complete Monty Python's Flying Circus: All the	Graham Chapman/John Cleese/Terry Gilliam/Eric Idle/Terry Jon	Pantheon	eng	100.00
1411677315	A Million Little Pieces of Feces	Python Bonkers	Lulu.com	en-US	100.00

Search Books

In the “Prev Orders” functionality, a customer will be able to view the previous orders they have made and also view the status of their previous orders. A status “open” means the books are yet to be delivered and a status “ “ means that order has been delivered and therefore that transaction is closed:

Kuraloviyan Book Store

SearchBook

Prev Orders

Recomm

Purchase

CustProfile

Logout

Books Order Status Details

Last name	isbn	Book Title	Transaction ID	Date	Status	Quantity	Price	Amount
senthilnathan	0131492020	Java: An Introduction to Problem Solving and Progr	1	April 26, 2021	open	5	100.00	500.00
senthilnathan	013237045X	Data Structures and Abstractions with Java	1	April 26, 2021	open	20	100.00	2000.00
senthilnathan	0471738840	Data Structures and Algorithms in Java	1	April 26, 2021	open	10	100.00	1000.00
senthilnathan	3822840858	1000 Record Covers	1	April 26, 2021	open	50	100.00	5000.00
senthilnathan	0312306180	Jenny and the Jaws of Life: Short Stories	2	April 26, 2021	open	3	100.00	300.00
senthilnathan	0441013252	The Outlaws of Sherwood	2	April 26, 2021	open	5	100.00	500.00
senthilnathan	0451219597	Jackdaws	2	April 26, 2021	open	20	100.00	2000.00
senthilnathan	0060730676	Buddhism: A Concise Introduction	11	April 29, 2021	open	5	100.00	500.00
senthilnathan	0060745908	Wicked: The Life and Times of the Wicked Witch of	11	April 29, 2021	open	6	100.00	600.00
senthilnathan	0060747226	Son of a Witch (The Wicked Years #2)	11	April 29, 2021	open	8	100.00	800.00

In the “Recomm” functionality, a customer would be able to see the Books recommended to them based on their previous orders. The functionality finds the latest book the customer has ordered and recommends all the books the same author has written except the book the customer previously ordered:

Kuraloviyan Book Store

SearchBook

Prev Orders

Recomm

Purchase

CustProfile

Logout

Books Recommendations

Last name	isbn	Book Author	Book Title
senthilnathan	0060753102	Robin McKinley	Beauty
senthilnathan	0060988649	Gregory Maguire/Douglas Smith	Lost
senthilnathan	0064404773	Robin McKinley	Beauty: A Retelling of the Story of Beauty and the
senthilnathan	0142302333	Robin McKinley	A Robin McKinley Collection: Spindle's End The He
senthilnathan	0321468937	Walter J. Savitch	Absolute C++
senthilnathan	0330332651	Ken Follett	A Dangerous Fortune
senthilnathan	0330345044	Ken Follett	Paper Money
senthilnathan	0441004997	Robin McKinley	The Hero and the Crown (Damar #2)
senthilnathan	0441005837	Robin McKinley	Rose Daughter
senthilnathan	0441012396	Robin McKinley	Deerskin
senthilnathan	0451207521	Ken Follett	Jackdaws
senthilnathan	0451208706	Ken Follett	The Man From St. Petersburg
senthilnathan	0451211472	Ken Follett	Night Over Water
senthilnathan	0515138819	Robin McKinley	Sunshine
senthilnathan	0552548227	Robin McKinley	Spindle's End
senthilnathan	0698119606	Robin McKinley	The Door in the Hedge
senthilnathan	0862031435	Robin McKinley	Beauty

In the Purchase Functionality, a customer would be able to search for books and add multiple different books of different quantities in their shopping cart.

First, they can search for the book in the search bar as seen below:

Kuraloviyen Book Store

SearchBook

Prev Orders

Recomm

Purchase

CustProfile

Logout

Search Books to Purchase

Search Book title or Author or Publisher or Subject

pyhton

display cart

They can then enter the quantity for the books they wish to purchase and then click the “add to cart” option:

Kuraloviyan Book Store

[SearchBook](#) [Prev Orders](#) [Recomm](#) [Purchase](#) [CustProfile](#) [Logout](#)

Book Details

isbn	Book Title	Author name	Publisher name	Language	Price	Order Quantity
0312311451	The Pythons Autobiography by The Pythons	Graham Chapman/John Cleese/Terry Gilliam/Eric Idle/Terry Jon	St. Martin's Griffin	eng	100.00	0
0679726470	The Complete Monty Python's Flying Circus: All the	Graham Chapman/Eric Idle/Terry Gilliam/John Cleese/Terry Jon	Pantheon	eng	100.00	10
0679726489	The Complete Monty Python's Flying Circus: All the	Graham Chapman/John Cleese/Terry Gilliam/Eric Idle/Terry Jon	Pantheon	eng	100.00	0
1411677315	A Million Little Pieces of Feces	Python Bonkers	Lulu.com	en-US	100.00	0

[Search Books](#) [add to cart](#)

They are then redirected back to search bar so they can search and order for more books. If they have finished adding books to the cart and wish to view their cart list and complete their order, they can click the “display cart” option:

Kuraloviyan Book Store

[SearchBook](#) [Prev Orders](#) [Recomm](#) [Purchase](#) [CustProfile](#) [Logout](#)

Shopping Cart Details

isbn	Book Title	Quantity	Price	NewQty
0679726470	The	10	100.00	10
013237045X	Data	40	100.00	40
0534371094	programming.java:	20	100.00	20
0060098910	The	5	100.00	5
0141439564	Great	100	100.00	100

[Search Books](#) [Purchase](#) [Update](#)

They can now view the Shopping Cart Details page and if they wish to delete one of their orders or change the quantities in one of their orders, they can simply change the quantity under the NewQty attribute and click the “Update” option (if they wish to delete one of the orders in their list, they can simply change the quantity of that order to 0 and then click update):

Kuraloviyan Book Store - [SearchBook](#) [Prev Orders](#) [Recomm](#) [Purchase](#) [CustProfile](#) [Logout](#)

Shopping Cart Details

isbn	Book Title	Quantity	Price	NewQty
0099429799	Atonement	50	100.00	50
067176991X	Mystery	100	100.00	100
0310929555	Study	2	100.00	0

[Search Books](#) [Purchase](#) [Update](#)

They can now see their updated shopping cart and then proceed to purchase their order by clicking the “Purchase” option:

Kuraloviyan Book Store - [SearchBook](#) [Prev Orders](#) [Recomm](#) [Purchase](#) [CustProfile](#) [Logout](#)

Shopping Cart Details

isbn	Book Title	Quantity	Price	NewQty
0099429799	Atonement	50	100.00	50
067176991X	Mystery	100	100.00	100

[Search Books](#) [Purchase](#) [Update](#)

In the “searchComments” functionality, a customer can search for a book based on the title or ISBN number and view the comments for that particular book by other customers:

Kuraloviyan Book Store

SearchBook

SearchComments

Prev Orders

Recomm

Purchase

CustProfile

Logout

Search Page

Search Book title or isbn

Kuraloviyan Book Store

SearchBook

SearchComments

Prev Orders

Recomm

Purchase

CustProfile

Logout

Book comments Details

Last name	isbn	Book Title	Comments	Feedback Date
senithilnathan	0131492020	Java: An Introduction to Problem Solving and Progr	Amazing book with great plot	April 29, 2021
senithilnathan	013237045X	Data Structures and Abstractions with Java	Amazing book with great plot	April 29, 2021
senithilnathan	0471738840	Data Structures and Algorithms in Java	bad book with boring plot	April 29, 2021

Search comments

In the “CustProfile” option, a customer can review other customers’ basic information, some of their comments, and the number of trusters/non-trusters through the profile page. He/she can also mark “trusted” or “not-trusted” on that page as well:

Customers

kural002 kuraloviyan senthilnathan	kural002
kural003 kuraloviyan senthilnathan	kural003
kural004 kural004 senthilnathan	kural004
kural005 kural005 senthilnathan5	kural005
kural006 kural006 senthil006	kural006
kural007 kural007 kural007	kural007
kural008 kural008 kural008	kural008

A Customer cannot view their own profile and therefore would not be able to mark their own comments usefulness ratings and would also not be able to mark their own Trust recordings:

Kuraloviyan Book Store

[SearchBook](#)
[Prev Orders](#)
[Recomm](#)
[Purchase](#)
[CustProfile](#)
[Logout](#)

Comments Details

Customer Name : senthilmnathan

isbn	Book Title	Book Score	Comments Text	Feedback Date	Others Rated	Usefulness
0131492020	Java: An Introduction to Problem Solving and Progr	2	Amazing book with great plot	April 29, 2021	Okay	Okay ▾
013237045X	Data Structures and Abstractions with Java	2	Amazing book with great plot	April 29, 2021	Good	Good ▾
0471738840	Data Structures and Algorithms in Java	2	bad book with boring plot	April 29, 2021	Good	Bad ▾
0312306180	Jenny and the Jaws of Life: Short Stories	2	Slow pacing and horrible plot	April 29, 2021	Good	Okay ▾

user legitimacy

☐ Not Trusted
 ☒ Trusted

Submit

When a customer has completed using their account, they can log out by simply clicking the “logout” option allowing others to access their own accounts:

Kuraloviyan Book Store

[Manager Login](#)
[Login](#)
[Register](#)

you have nice day...

When Manager wishes to login, they must use the manager login page:

Manager Login

User Login ID: User password:

Manager Login

Cancel

In the SearchBook functionality, the manager can search for a book title or an author or a particular publisher or even a favorite subject:

In the “BooksAdmin” option, the manager can perform book administration functionalities like Adding a new book using the “create new” option, change existing book details like the Stock level by using the “edit” option or can also delete a book:

BOOKS

create new

The Prophet Kahlil Gibran/جبران خليل جبران/Jihad El Rupa & Co eng computer 100.00

edit

delete

Sam And The Firefly (Beginner Books) P.D. Eastman Collins eng computer 100.00

edit

delete

After the Funeral Agatha Christie HarperCollins eng computer 100.00

edit

delete

Sleeping Murder (Miss Marple #13) Agatha Christie Dodd Mead; 1st edition (September 1976) eng computer 100.00

edit

delete

The One Tree (The Second Chronicles of Thomas Cove Stephen R. Donaldson Fontana en-GB computer 100.00

edit

delete

The Great And Secret Show (Book of the Art #1) Clive Barker Fontana Books eng computer 100.00

edit

delete

Rage of Angels Sidney Sheldon HarperCollins Publishers eng computer 100.00

edit

delete

In the “create new” option, Managers can add a new book by filling all the details of the book as shown below:

Kuraloviyen Book Store ▾ SearchBook BooksAdmin PendingOrders BestCustomer BestBooks New Manager ManagerLogout

All fields are required

Isbn:

Book title:

Author name:

Language:

Publisher name:

Edition_number:

Size:

Size:

Price:

Keyword:

Subject:

No_purchased_books:

No_sold_books:

post

In the “edit” option, a Manager would be able to change the book details like updating the price or the stock level:

Size:

Price:

Keyword:

Subject:

No_purchased_books:

No_sold_books:

[post](#)

In the “delete” option, a manager can use it to delete a book if that book is no longer being sold or is discontinued.

In the PendingOrders functionality, a manager can view all orders whose status are still “open” and not “closed”:

Kuraloviyan Book Store - SearchBook BooksAdmin PendingOrders BestCustomer BestBooks New Manager ManagerLogout								
Books Order Status Details								
Last name	isbn	Book Title	Transaction ID	Date	Status	Quantity	Price	Amount
senthilmnathan	0131492020	Java: An Introduction to Problem Solving and Progr	1	April 26, 2021	open	5	100.00	500.00
senthilmnathan	013237045X	Data Structures and Abstractions with Java	1	April 26, 2021	open	20	100.00	2000.00
senthilmnathan	0471738840	Data Structures and Algorithms in Java	1	April 26, 2021	open	10	100.00	1000.00
senthilmnathan	3822840858	1000 Record Covers	1	April 26, 2021	open	50	100.00	5000.00
senthilmnathan	0312306180	Jenny and the Jaws of Life: Short Stories	2	April 26, 2021	open	3	100.00	300.00
senthilmnathan	0441013252	The Outlaws of Sherwood	2	April 26, 2021	open	5	100.00	500.00
senthilmnathan	0451219597	Jackdaws	2	April 26, 2021	open	20	100.00	2000.00
senthilmnathan	0007179812	Microserfs	6	April 29, 2021	open	10	100.00	1000.00
senthilmnathan	0060080833	After	6	April 29, 2021	open	20	100.00	2000.00
senthilmnathan	0060507276	Household Saints	6	April 29, 2021	open	5	100.00	500.00
senthilmnathan	006058081X	Cookies: Bite-Size Life Lessons	6	April 29, 2021	open	6	100.00	600.00
senthilmnathan	0061127760	Charlotte's Web	6	April 29, 2021	open	50	100.00	5000.00
senthilmnathan	0060011572	Animal Liberation	7	April 29, 2021	open	3	100.00	300.00
senthilmnathan	0060084383	The Modern Mind: An Intellectual History of the 20	7	April 29, 2021	open	5	100.00	500.00
senthilmnathan	0060521228	The Whipping Boy	7	April 29, 2021	open	6	100.00	600.00
senthilmnathan	0060826584	Oracle Bones: A Journey Between China's Past and P	7	April 29, 2021	open	8	100.00	800.00
senthilmnathan	0060882603	The Annotated Charlotte's Web	7	April 29, 2021	open	10	100.00	1000.00
senthilmnathan5	0060519568	Touching Darkness (Midnighters #2)	8	April 29, 2021	open	8	100.00	800.00
senthilmnathan5	0060519592	Blue Noon (Midnighters #3)	8	April 29, 2021	open	6	100.00	600.00
senthilmnathan5	006109398X	Treasure Box	8	April 29, 2021	open	5	100.00	500.00
senthilmnathan5	0061093998	Homebody	8	April 29, 2021	open	20	100.00	2000.00
senthilmnathan5	0099690608	The Abyss	8	April 29, 2021	open	12	100.00	1200.00
senthilmnathan5	0140436588	Ivanhoe	8	April 29, 2021	open	5	100.00	500.00

Only a manager can add a new manager and can use the “New Manager” functionality to add a new manager to the bookstore:

Kuraloviyan Book Store [SearchBook](#) [BooksAdmin](#) [PendingOrders](#) [BestCustomer](#) [BestBooks](#) [New Manager](#) [ManagerLogout](#)

New manager Information

Login ID:

First name:

Last name:

Mobile number:

Email id:

User password:

[Register](#)

The constraints to registration are:

- (f) If the Login ID already exists, the system would ask to try another login ID.
- (g) Mobile Number would only accept numbers and no other characters.
- (h) Email ID needs to contain the @ character.
- (i) The password length should not be less than 8 characters in length, and it is hashed while typing.
- (j) No field can be left blank.

In the BestCustomer functionality, the manager can review statistics like the top 5 Customers based on the Invoice Amount:

Kuraloviyan Book Store [SearchBook](#) [BooksAdmin](#) [PendingOrders](#) [BestCustomer](#) [BestBooks](#) [New Manager](#) [ManagerLogout](#)

Top 5 Cusotmers by Invoice Amount

customer ID	Last name	Total Invoice Amount
kural002	senthilnathan	28200.00
kural003	senthilnathan	9100.00
kural006	senthil006	6000.00
kural005	senthilnathan5	5600.00
kural007	kural007	5000.00

In the BestBooks functionality, the manager can review statistics like the top 5 Books based on the Sales quantity for that year:

Kuraloviyan Book Store

SearchBook

BooksAdmin

PendingOrders

BestCustomer

BestBooks

New Manager

ManagerLogout

Top 5 Books by Sales quantity of this Year

isbn	Book Title	Total Quantity Sales	Total sales amount
067176991X	Mystery Walk	100	10000.00
0061127760	Charlotte's Web	50	5000.00
0099429799	Atonement	50	5000.00
3822840858	1000 Record Covers	50	5000.00
0843949740	The Reluctant Reformer	30	3000.00

If the manager has completed his/her work and wishes to log out , they can choose the “ManagerLogout” option:

Kuraloviyan Book Store	Manager Login	Login	Register
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you have nice day...